i. **Dissolved in plasma** About 7 – 10% of CO_2 is transported in a dissolved form in the plasma.

ii. **Bound to haemoglobin** About 20 – 25% of dissolved CO_2 is bound and carried in the RBCs as carbaminohaemoglobin (Hb CO_2)

$$CO_2 + Hb \rightleftharpoons Hb\,CO_2$$

iii. **As bicarbonate ions in plasma** about 70% of CO_2 is transported as bicarbonate ions

This is influenced by pCO_2 and the degree of haemoglobin oxygenation. RBCs contain a high concentration of the enzyme, carbonic anhydrase, whereas small amounts of carbonic anhydrase is present in the plasma.

At the tissues the pCO_2 is high due to catabolism and diffuses into the blood to form HCO_3^- and H^+ ions. When CO_2 diffuses into the RBCs, it combines with water forming carbonic acid (H_2CO_3) catalyzed by carbonic anhydrase. Carbonic acid is unstable and dissociates into hydrogen and bicarbonate ions.

Carbonic anhydrase facilitates the reaction in both directions.

$$CO_2 + H_2O \underset{\text{carbonic anhydrase}}{\rightleftharpoons} H_2CO_3 \underset{\text{carbonic anhydrase}}{\rightleftharpoons} HCO_3^- + H^+$$

The HCO_3^- moves quickly from the RBCs into the plasma, where it is carried to the lungs. At the alveolar site where pCO_2 is low, the reaction is reversed leading to the formation of CO_2 and water. Thus CO_2 trapped as HCO_3^- at the tissue level it is transported to the alveoli and released out as CO_2. Every 100mL of deoxygenated blood delivers 4mL of CO_2 to the alveoli for elimination.

6.6 Regulation of Respiration

A specialised respiratory centre present in the medulla oblongata of the hind brain called **respiratory rhythm centre** is responsible for this regulation. **Pneumotaxic centre** present in pons varoli region of the brain moderates the function of the respiratory rhythm centre to ensure normal breathing. The chemosensitive area found close to the rhythm centre is highly sensitive to CO_2 and H^+. And H^+ are eliminated out by respiratory process. Receptors associated with the aortic arch and carotid artery send necessary signals to the rhythm centre for remedial action. The role of O_2 is insignificant in the regulation of respiratory rhythm.

Particulate matter PM 2.5 in the air is increasing day by day which causes respiratory illness. Central Pollution Control Board (CPCB) reports that the quality of air is not good due to soot and smoke. So some cities in India are using CNG (Compressed Natural Gas) as fuel.

6.7 Problems in Oxygen transport

When a person travels quickly from sea level to elevations above 8000ft, where the atmospheric pressure and partial pressure of oxygen are lowered, the individual responds with symptoms of **acute mountain sickness** (AMS)– headache, shortness of breath, nausea and dizziness due to poor binding of O_2 with haemoglobin. When the person moves on a long–term basis to mountains from sea level is the body begins to

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